

CSE Alpha Engine — Part II

Decision Layer, Live Data Infrastructure & Agent Supervision

Version: 1.0 — companion to *CSE Alpha Engine Architecture v1.0* **Covers:** the buy-price engine, the human-in-the-loop decision record, the Claude supervisory agent, and the always-on data service.

0. What Changed From Part I

Part I defined the models. Part II defines **how the models become a number you can act on, and how the system learns from your decisions.**

Three requirements from you, restated as design constraints:

Your requirement	Design consequence
"6+ months, catch it before it booms, know when to sell"	Horizon locked at 6–18 months . This kills any intraday logic and sets the entire refresh cadence.
"Not one verdict — show me the numbers under each model, I decide"	The app has no BUY button and no single recommendation . It has an evidence panel and a decision-capture form. The models never conclude; they present.
"Exact price below which I should buy, practical vs current market"	Needs a fair-value triangulation engine with a derived margin of safety and an order-book-aware entry ladder. This is Section 2 — the most important new component.
"An agent that reviews what happened and improves the system"	A three-tier supervisory agent with hard authority limits. This is the highest-risk component in the whole build. Section 4 is mostly guardrails, deliberately.
"Running on a server 24/7, always fetching latest"	Section 5. Includes an honest reframing of what "24/7" should actually mean for a 6–18 month strategy.

1. Horizon Design — Locking the Clock

Everything in the system must run on one clock. Mixing horizons is how quant systems quietly destroy themselves.

Component	Horizon	Refresh cadence
M4 macro regime	3–12 month view	Monthly + event-triggered
M2 fundamentals	6–18 month view	On earnings release + monthly re-rank
M3 timing	1–6 month entry window	Daily
M1 factor certification	3-year estimation window	Weekly
Holding period target	6–18 months	—
Price monitoring	—	Every 5 min during market hours

Note the asymmetry: **prices are watched constantly, models are not re-estimated constantly.** Re-running an ARDL macro model every five minutes is theatre — the underlying data is monthly. What runs frequently is one cheap question: *has any watchlist name crossed its buy-below price?* That's a comparison, not a computation.

Target trade profile: 8–15 positions, 15–25 round trips a year, average hold 9 months. At ~2.24% in fees plus spread, 20 round trips costs you roughly 5–7% of capital annually in friction. That is your baseline drag, and it is why the 8% expected-return floor from Part I exists.

2. The Buy-Price Engine — “Below what price should I buy this?”

This is the component that turns analysis into an executable number. It runs in four stages.

Stage 1 — Three independent fair-value anchors

Never trust one valuation method. Compute three, then look at their spread — the *disagreement between them is itself information.*

Anchor A — Multiple reversion (from M2)

Normalised EBIT = 3-year average EBIT, adjusted for one-offs
Target multiple = MIN(sector median EV/EBIT, stock's own 5-year median EV/EBIT)
Implied EV = Normalised EBIT × Target multiple
FV_A = (Implied EV - net debt - minority interests) ÷ shares outstan

Using the lower of sector and own-history multiple is deliberately conservative — it stops you assuming a structurally cheap company will suddenly re-rate to the sector.

Anchor B — Asset / sum-of-the-parts (from M2)

For conglomerates: $\Sigma(\text{listed subsidiaries at market value} \times \text{ownership \%})$
+ $\Sigma(\text{unlisted subsidiaries at sector EV/EBITDA})$
+ investment property at hard book
- holdco net debt
- holding company discount (15-30%, calibrated to that company's own historical average discount)
÷ shares outstanding
For asset-heavy: hard book value per share × justified P/B
where justified P/B = $(\text{ROE} - g) \div (\text{Ke} - g)$

Anchor C — Required-return model (from M1 + M4 — this is where the factor work pays off)

$$\text{Ke} = \text{Rf} + \beta_{\text{MKT}} \cdot \text{E}[\text{MRP}|\text{regime}] + \beta_{\text{SMB}} \cdot \text{E}[\text{SMB}|\text{regime}] + \beta_{\text{HML}} \cdot \text{E}[\text{HML}|\text{regime}] + \beta_{\text{MOM}} \cdot \text{E}[\text{MOM}|\text{regime}] + \beta_{\text{LIQ}} \cdot \text{E}[\text{LIQ}|\text{regime}]$$

where Rf = current 364-day T-bill yield

$\text{E}[\cdot|\text{regime}]$ = M4's regime-conditional factor premia

$$\text{FV}_C = \Sigma (\text{forecast FCFE}_t \div (1+\text{Ke})^t) + \text{terminal value}$$

This is the elegant part of the architecture: **M1's Dimson-corrected factor loadings and M4's regime forecast combine to produce a discount rate, and the discount rate produces a price.** Your macro view and your risk model are not decoration — they land directly in the buy price. When M4 flips to Risk-Off, Ke rises, and every fair value in the system falls automatically.

Blended fair value:

$$\text{FV} = w_A \cdot \text{FV}_A + w_B \cdot \text{FV}_B + w_C \cdot \text{FV}_C$$

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Company type	w_A	w_B	w_C
Operating company, stable earnings	0.45	0.15	0.40
Conglomerate / holding company	0.20	0.55	0.25
Bank / finance	0.25	0.50 (justified P/B)	0.25
Cyclical (hotels, plantations, construction)	0.35	0.35	0.30

Stage 2 — Derive the margin of safety (don't guess it)

The margin of safety is **computed**, not a habit. It is the sum of five components:

$$\text{MoS} = \text{MoS_base} + \text{MoS_dispersion} + \text{MoS_liquidity} + \text{MoS_regime} + \text{MoS_quality}$$

Component	Rule	Range
Base	Covers round-trip friction + minimum required edge	10% (fixed)
Dispersion	$(\text{max FV} - \text{min FV}) \div \text{mean FV} \times 0.5$ — the more your three anchors disagree, the less you know	0– 15%
Liquidity	Amihud percentile mapped: top quartile liquid → 0%, bottom quartile → 10%	0– 10%
Regime	Risk-On 0% · Transition +5% · Risk-Off +12%	0– 12%
Quality	$(70 - \text{M2_pillarD_score}) \div 4$, floored at 0 — pay less for opaque accounts	0–8%

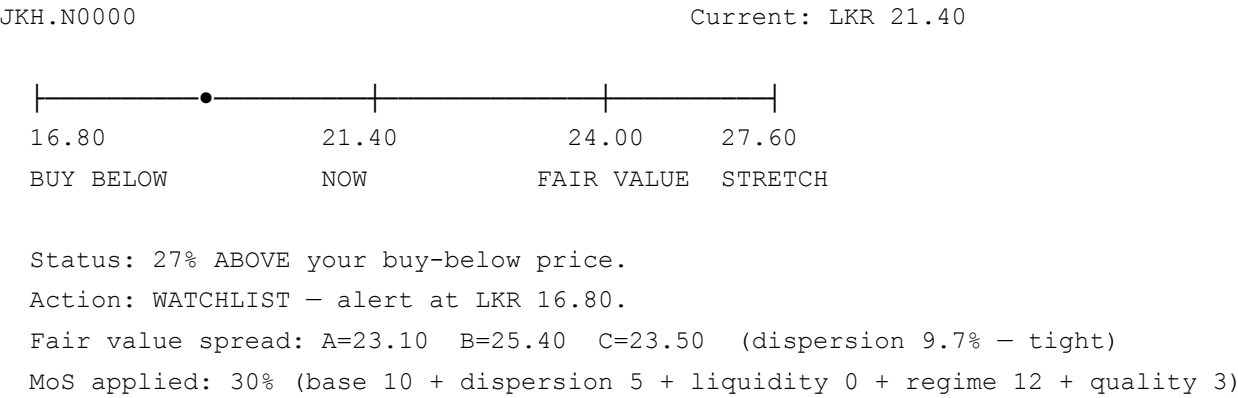
Total MoS lands between 10% and 55%. A high number is not a bug — it is the system telling you it doesn't understand this company well enough to pay up.

Stage 3 — The three prices you actually need

BUY BELOW	= $\text{FV} \times (1 - \text{MoS})$	← your limit price ceiling
FAIR VALUE	= FV	← the anchor
TRIM ABOVE	= $\text{FV} \times 1.00$	← start scaling out
STRETCH EXIT	= $\text{FV} \times 1.15$	← only if M3 momentum still positive

The UI shows this as a single horizontal bar with the **current market price marked on it**,

and the gap stated in plain language:



(Illustrative numbers only — populate from live estimation.)

This is the answer to “at what price should I buy.” It is derived, auditable, and it moves when the macro regime moves.

Stage 4 — Practical execution vs the actual order book

A theoretical limit price is useless if the book is 400 shares deep. The engine adds:

Field	Computation
best_ask / depth_at_ask	From live detailedTrades / order book
entry_ladder	Tranche 1: 40% at MIN(buy_below, best_ask) · Tranche 2: 35% at buy_below × 0.98 · Tranche 3: 25% at buy_below × 0.95
days_to_build	target_position_value ÷ (0.15 × ADV_20d)
est_slippage	Amihud-scaled impact for your order size
all_in_entry_cost	limit price × (1 + 0.0112) + est_slippage
breakeven_price	Entry price grossed up for both-way costs — display this always. Seeing “you need +2.3% just to break even” changes behaviour.

If days_to_build > 10 , the UI flags it: *good company, wrong size for you.*

3. The Decision Layer — You Decide, the System Records

3.1 Design principle: surface disagreement, not consensus

The app's job is not to conclude. It is to **put the four models side by side and make their disagreement impossible to miss** — because that is exactly where your judgement adds value. When all four models align, the decision is easy and you barely need the app. When M2 says "cheap and clean" while M3 says "still falling" and M4 says "Risk-Off," that's the moment worth thinking hard about.

The Decision Card — one screen per candidate, no verdict anywhere on it:

SAMP.N0000	Current LKR 78.50	Buy below 61.20
M1 CARHART	α +6.2% p.a. (t=1.81) $R^2=0.42$ β : MKT 0.94 SMB 0.31 HML 0.66 MOM -0.12 → Genuine alpha; value-tilted; not a momentum bet. Residual vol 24%.	
M2 FUNDAMENTALS	Score 81 / 100 ▲ Value 88 · Quality 74 · Safety 79 · Integrity 83 · F-Score 7/9 Flags: none	
M3 TIMING	Score 44 / 100 ▼ Branch: NEUTRAL 52wk-high proximity 0.71 · Resid mom -3% → Verdict: WAIT FOR CONFIRMATION	
M4 MACRO	Regime: TRANSITION (P=0.61) Banks sector fit: NEUTRAL Rate path favourable / FX pressure adverse	
▲ MODELS DISAGREE — fundamentals strong, timing weak. Historical base rate for this pattern: 54% hit rate, avg +11.2% at 6m, but 3.1 month average drawdown first.		
YOUR DECISION: [Buy] [Watchlist] [Pass] [Buy partial] Size: ____% Limit: ____ Conviction: [1-5] Reasoning: _____ What would prove me wrong? _____		

Two fields there matter more than they look. **"Conviction 1–5"** lets you later test whether your own confidence is calibrated — most people's isn't, and finding out is worth a lot.

"What would prove me wrong?" is the falsification field: written *before* the trade, it makes post-mortems honest instead of retrospective storytelling.

3.2 Decision record schema

```
decisions (  
  decision_id, ticker, timestamp, config_hash,  
  action,                                -- buy / watchlist / pass / partial / sell / trim  
  size_pct, limit_price, conviction_1_5,  
  reasoning_text, falsification_text,  
  -- full model state frozen at decision time:  
  m1_alpha, m1_tstat, m1_betas_json, m1_resid_vol,  
  m2_score, m2_pillars_json, m2_flags_json,  
  m3_score, m3_branch, m3_verdict, m3_signals_json,  
  m4_regime, m4_prob, m4_sector_fit,  
  fv_a, fv_b, fv_c, fv_blended, mos_applied, buy_below,  
  market_price_at_decision,  
  agreement_score,                       -- how aligned the four models were  
  override_flag                          -- did the human go against the composite?  
)  
  
outcomes (  
  decision_id, exit_date, exit_price, exit_trigger,  
  gross_return, net_return, holding_days,  
  max_adverse_excursion, max_favourable_excursion,  
  attribution_json                       -- model / execution / macro / luck  
)
```

Freezing the full model state at decision time is what makes learning possible. Without it you can never answer “was the model wrong, or was I?”

4. The Claude Supervisory Agent

This is the component you’re most excited about and the one most likely to lose you money if built carelessly. An LLM that retunes a trading system based on recent results will, left unconstrained, chase noise and change parameters at exactly the wrong moments — after losses, which is when reversion is most likely. So the agent gets **three tiers of authority, and it earns its way up.**

Tier A — Narrator (*read-only · deploy from day one · zero risk*)

Function	Output
Pre-trade brief	Plain-language synthesis of all four models for a candidate, explicitly naming where they conflict and what each conflict historically implied

Regime commentary	Translates M4's ARDL/IRF output into readable prose: what changed, why, what it implies for exposure
Disclosure triage	Reads new CSE announcements, classifies (earnings / corporate action / related-party / governance / capital raise), flags anything touching a held or watchlist name within minutes
Anomaly explanation	"GREG.N0000 moved +14% on 4x normal volume with no announcement" — surfaces, doesn't interpret
Post-mortem draft	On every closed position: what the models said, what you said, what happened, and a first-pass attribution

Tier A is where **most of the agent's real value lives**. LLMs are genuinely excellent at synthesis, translation and pattern-narration across messy text. Lean on this hard.

Tier B — Diagnostician (*read + propose · from month 3 · proposals only*)

Reviews the accumulated `decisions` and `outcomes` tables and answers questions you can't easily see yourself:

- Which model has the highest rank IC on *your actual closed trades*, versus in backtest?
- Do your overrides add or destroy value? (*Compare returns on trades where you went against the composite vs. with it. This is the single most valuable analysis the agent can run.*)
- Is your conviction calibrated? Do 5s outperform 3s?
- Where do losses cluster — which sector, which score band, which regime, which liquidity bucket?
- Are entries systematically early or late? (Use max adverse excursion: consistently large MAE before profit means your entry ladder is too aggressive.)
- Is any model decaying? Rolling IC vs. backtest IC.

Output format: a proposal ticket, never a change.

PROPOSAL #14

Observation: 12 of your 15 losing trades since Jan had M2 Integrity score between 50 and 65. Winners averaged 79.

Sample: 38 closed positions, 14 months.

Hypothesis: The Pillar D veto threshold is too permissive.

Proposed: Raise hard-veto floor from 45 to 60.

Backtest: [Tier C runs this]
 Risk: Reduces universe by ~11 names, may cut valid opportunities.
 Status: AWAITING HUMAN REVIEW

Tier C — Optimiser (*propose + auto-validate · month 12+ · still cannot deploy*)

Can execute a proposal through the **full Part I validation harness** — purged walk-forward, cost model, Deflated Sharpe, baseline comparison, regime-by-regime breakdown, parameter sensitivity sweep — and present the results. **It cannot merge.** You merge.

4.1 Hard guardrails — non-negotiable

#	Guardrail	Why
1	Minimum sample: 30 closed positions AND 12 months before any proposal is even generated	Below this, every “pattern” is noise. This rule alone prevents most of the damage.
2	Parameter freeze during drawdown > 15%	Retuning after losses is the classic wealth-destroying reflex. The agent is explicitly forbidden from proposing changes while you’re hurting.
3	Change budget: 1 parameter change per quarter, maximum	Rate-limiting forces prioritisation and keeps the system attributable. Ten simultaneous changes teach you nothing.
4	Every proposal passes the same bar as an original model	Purged walk-forward, DSR > 0, must beat the simple baseline. No exceptions for agent-authored changes.
5	The cost model, liquidity caps, position limits and Pillar D veto are CONSTITUTIONAL — the agent may never propose changes to them	These are your survival constraints, not optimisable parameters. An optimiser will always want to relax them because relaxing them improves backtest returns.
6	Full config versioning — every model config is git-tracked with a hash; every signal and decision is stamped with the hash that produced it; one-click rollback	Without this you cannot tell which version generated which result.
7	The agent is scored too	Track the realised performance of its accepted proposals. If its proposals underperform the unchanged system over

		a year, demote it to Tier A.
8	Agent never has write access to the database or order flow	It writes tickets. Humans write parameters. Nothing writes orders.

4.2 The honest assessment

Claude will be very good at Tier A and genuinely useful at Tier B. **Tier C is where to be slow.** The reason isn't capability — it's sample size. With 20–30 trades a year, you accumulate statistically meaningful evidence about your own system at a rate of roughly one solid conclusion per 18 months. No amount of model intelligence compresses that. The agent's most valuable contribution over the first two years will be *making you honest about your own decisions*, not tuning coefficients.

5. Always-On Data Infrastructure

5.1 Reality check on "24/7"

The CSE trades roughly **09:30–14:30 local, five days a week** — about 25 hours out of 168. Your strategy horizon is 6–18 months. So what actually runs continuously is the *scheduler*, *the monitor* and *the overnight batch* — not a firehose of redundant polling. Polling a closed exchange every minute burns rate limit and buys nothing. Here's the correct cadence:

Job	Schedule	Purpose
<code>marketStatus</code> check	Every 5 min, always	Cheap heartbeat; gates everything else
Price snapshot (<code>tradeSummary</code> , <code>todaySharePrice</code>)	Every 5 min, market hours only	Live prices for the gap-to-entry display
Buy-below trigger check	Every 5 min, market hours	<i>The only intraday computation that matters:</i> has any watchlist name crossed its limit? → push alert
Order book (<code>detailedTrades</code>)	Every 15 min, watchlist names only	Depth for the entry ladder
Announcements		Corporate actions and

(approvedAnnouncement , new listings, buy-in board)	Every 15 min, 08:00–18:00	earnings land here — highest-value feed in the system
EOD snapshot + adjustment	15:00 daily	Close, volume, corporate-action adjustment, reconciliation test
Nightly model batch	20:00 daily	Recompute M2/M3 scores, fair values, buy-below prices, factor returns
M1 factor regressions	Weekly, Sunday 02:00	3-year rolling window re-estimation
CBSL macro poll	Release-calendar driven — T-bill auction Wednesdays, CCPI end-of-month, policy decisions ~8/yr	Event-driven, not continuous
M4 macro re-estimation	Monthly, + immediately on policy decision or IMF milestone	ARDL/VECM/regime refresh
Full validation re-run	Quarterly	Drift and decay check

5.2 The cse.lk data path — and its risk

The CSE web portal is backed by undocumented JSON endpoints under

`https://www.cse.lk/api/` — `marketStatus` , `marketSummery` , `tradeSummary` , `todaySharePrice` , `companyInfoSummery` , `detailedTrades` , `dailyMarketSummery` , `aspiData` , `snpData` , `topGainers` , `topLooses` , and announcement endpoints among them.

These are reverse-engineered rather than officially documented — no official documentation exists, and they require no authentication but do warrant sensible self-imposed rate limiting.

This is the single biggest operational fragility in your build. Treat it accordingly:

Risk	Mitigation
Endpoint changes or disappears without notice	Schema validation on every response; contract tests in CI; alert on shape change, not just on error

Rate limiting or IP block	Conservative pacing (≥ 2 s between calls), exponential backoff, circuit breaker, identifying user-agent, never parallel-hammer
Silent stale data	Every record carries <code>fetches_at</code> ; UI shows a stale-data banner if the last successful fetch is older than the expected interval; models refuse to emit new signals on stale inputs
Wrong data (not just missing)	Nightly cross-check against a second source — broker EOD file or the CSE mobile app feed. Discrepancy $> 0.5\%$ quarantines the ticker.
Terms of use	Personal-use, low-frequency access. If this ever scales beyond personal use, move to a licensed feed — ICE, for instance, <cite index="22-1"> distributes CSE Level 1 streaming and end-of-day data through its Consolidated Feed</cite>. Budget for this eventuality.

Design rule: a failed fetch must never silently produce a stale signal. Fail loudly, degrade visibly, and let the UI show yesterday's number clearly labelled as yesterday's.

5.3 Deployment

Nothing exotic is needed. Resist the urge to over-architect:

```

Single VPS (2 vCPU / 4GB — Hetzner, DigitalOcean, or similar)
├─ PostgreSQL + TimescaleDB    ← PIT store
├─ Redis                        ← computed scores, session cache
├─ FastAPI                     ← API for the React frontend
├─ APScheduler / Prefect       ← the job table above
├─ Worker process               ← model batch
└─ Caddy / nginx + TLS

Frontend: React static build (Vercel/Netlify free tier)
Alerts:   Telegram bot or email — push when a buy-below is crossed
          or an announcement hits a held name
Backups:  Nightly pg_dump to object storage, 30-day retention
Agent:    Claude API, called by the worker for Tier A/B tasks

```

Realistic running cost: **\$25–50/month** including the VPS, storage, and moderate Claude API usage. If a design decision pushes this toward hundreds a month, it's probably the wrong decision at this stage.

6. Dashboard — Home Screen Spec

One screen, answering four questions in order of importance:

1. WHAT'S THE WEATHER?
Regime: TRANSITION (P=0.61) ▼ from Risk-On
Recommended gross exposure: 65% Actual: 82% ▲
Earnings yield - 364d T-bill spread: [hero chart]
Next events: CBSL policy (12d) · CCPI (8d) · IMF (31d)
2. WHAT NEEDS MY ATTENTION TODAY?
• 2 watchlist names crossed buy-below
• 1 held name hit an exit trigger
• 3 new announcements on held names
• 1 data quality alert
3. WHERE AM I?
Portfolio: 9 positions · +14.2% YTD vs ASPI +6.8%
Portfolio factor exposure vs target [radar]
Hit rate (trailing 20): 61% · Win/loss: 2.1×
4. WHAT'S ON THE BOARD?
Top 10 by composite, each showing: current price, buy-below, gap %, four model scores, agreement flag

Sections 1 and 2 are what you check daily in two minutes. Sections 3 and 4 are for a weekly sit-down. **If the dashboard requires more than two minutes on a normal day, it is badly designed** — a 6–18 month strategy should not demand daily attention, and a UI that manufactures urgency will push you into overtrading, which is precisely how the cost structure kills you.

7. Revised Build Sequence

Phase	Adds	Weeks
1	PIT data spine, cse.lk ingestion, corporate actions, reconciliation, universe gates	3–4
2	Factor library, M2 baseline screen, buy-price engine (Stages 1–3) , basic screener UI	4–5

3	Scheduler + always-on service, alerting, dashboard sections 1–2, decision capture	2–3
4	M4 macro (ARDL/regime), regime dashboard, regime-conditional MoS	3–4
5	M1 Carhart + M3 timing, full fusion, complete Decision Card	3–4
6	Agent Tier A (narrator, disclosure triage, post-mortems)	1–2
7	Execution layer, backtest lab, monitoring, kill switches	3–4
8	(month 12+) Agent Tier B, then C — gated on 30 closed positions	—

Note the deliberate ordering: decision capture ships in Phase 3, before three of the four models exist. That’s intentional. Every decision you make without a recorded rationale is a data point you can never recover, and the agent’s future usefulness is a direct function of how early you start logging. Start recording decisions the day you can see a price and a score — even if the score is only M2.

8. Open Questions to Resolve Before Phase 1

1. **Capital base** — sets liquidity thresholds and universe size. LKR 5m and LKR 100m are different systems.
2. **Historical data depth** — how far back can you reconstruct point-in-time fundamentals? This caps your backtest and therefore your confidence. Realistically you may find 2015-onward is clean and earlier is patchy.
3. **Broker execution** — manual order entry, or does your broker offer an API? Determines whether entry ladders are executed by you or by the system.
4. **Tax treatment** — capital gains treatment on your CSE holdings, and your residency position, both affect net returns and optimal holding period. Worth resolving before the cost model is finalised, since it sits in the same equation.
5. **Second data source** — identify one now for the cross-check. A single unofficial endpoint is a single point of failure.

Companion to Part I. Together these two documents specify the complete system. Appendices (data dictionary, formula reference, sector mapping, parameter defaults) to be populated during Phase 1.

